

PracticeMidterm2_VA

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Math 340: Elementary Matrix and Linear Algebra

PRACTICE MIDTERM TWO

Name: key Section#: _____ NetID: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of five problems on six pages; the exam is four sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculator and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (4+4+5 points) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}$$

be four vectors from $\mathbb{R}^3$.

(a) Let $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a set of vectors. Is S_1 linearly independent?

$$\det \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 1 \\ 0 & 1 & 3 \end{bmatrix} = 1 \cdot (-3 - 1) - 2(6) = -4 - 12 = -16 \neq 0.$$

Yes!

(b) Let $S_2 = \{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be another set of vectors. What is the dimension of $\text{span } S_2$?

$$\begin{bmatrix} 2 & 0 & 4 \\ -1 & 1 & -1 \\ 1 & 3 & 5 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} -1 & 1 & -1 \\ 2 & 0 & 4 \\ 1 & 3 & 5 \end{bmatrix} \xrightarrow{\begin{matrix} 2r_1 + r_2 \rightarrow r_2 \\ r_1 + r_3 \rightarrow r_3 \end{matrix}} \begin{bmatrix} -1 & 1 & -1 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix}$$

$$\xrightarrow{-2r_2 + r_3 \rightarrow r_3} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}r_2 \rightarrow r_2} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_2 + r_1 \rightarrow r_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

basis: $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$

$\dim \text{span } S_2 = 2.$

(c) Let $V = \text{span } S$, where

$$S = \left\{ \overset{A_1}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}, \overset{A_2}{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}, \overset{A_3}{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}, \overset{A_4}{\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}} \right\}.$$

Find a basis for V as a subspace of $\mathbb{R}^{2 \times 2}$.

Soln A: Rewrite $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

$$\begin{array}{c} A_1 \quad A_2 \quad A_3 \quad A_4 \\ \left[\begin{array}{cccc} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 \end{array} \right] \xrightarrow{\text{row reduce}} \begin{array}{c} \text{pivot} \\ \left[\begin{array}{cccc} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

basis.

Use $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 $A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ as basis.

Soln B: Note that $A_4 = -A_1 + A_2$
 $A_3 = A_1 + A_2 \quad \left. \vphantom{\begin{matrix} A_4 = -A_1 + A_2 \\ A_3 = A_1 + A_2 \end{matrix}} \right\} \rightarrow \text{span}\{A_1, A_2\} = \text{span } S$
 and A_1, A_2 are linearly independent, since

$$A_2 \neq kA_1 \text{ so}$$

A_1, A_2 form a basis for $\text{span } S$.

2. (4+5 points)

(a) Let $B = (t-1, t^2-1, 2t+2)$ be an ordered basis for P_2 and $p = 3t^2 - 7t - 12$ be a polynomial in P_2 . Find $[p]_B$.

$\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ where $a(t-1) + b(t^2-1) + c(2t+2) = 3t^2 - 7t - 12$.

$$\textcircled{b}t^2 + \textcircled{a+2c}t + \textcircled{-a-b+2c} = \textcircled{3}t^2 - \textcircled{7}t - \textcircled{12}$$

$$[p]_B = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$$

$$\begin{aligned} b &= 3 & a+2c &= -7 \\ + -a-b+2c &= -12 \\ \hline -b+4c &= -19 \end{aligned}$$

$$4c = -19 + 3 = -16 \Rightarrow c = -4$$

$$a = 1$$

(b) Let $f: \mathbb{R}^3 \rightarrow P_1$ be a function defined by

$$f\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = (a-b)t + (c+a).$$

- Is f injective (one-to-one)?
- Is f surjective (onto)?
- Is f an isomorphism?

(i) No, $f\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) = (1-0)t + (1+1) = t+2$

$f\left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) = (2-1)t + 0+2 = t+2$ $\swarrow$ same!

(ii) Sure, for $a_1t + a_2$

$$f\left(\begin{bmatrix} 0 \\ -a_1 \\ a_2 \end{bmatrix}\right) = (0+a_1)t + 0+a_2 = a_1t + a_2 \checkmark$$

(iii) Not an isomorphism b/c it's not a bijection.

3. (5+5 points) Let

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -3 & 1 \\ 1 & -6 & -7 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & 2 \\ (c^2-2) & 0 & 4 \\ 2 & 1 & 3 \end{bmatrix}$$

be two 3×3 matrices.

(a) Determine the rank and the nullity of A .

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & -3 & 1 \\ 1 & -6 & -7 \end{bmatrix} \xrightarrow{\substack{-2r_1+r_2 \rightarrow r_2 \\ -r_1+r_3 \rightarrow r_3}} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -3 & -5 \\ 0 & -6 & -10 \end{bmatrix} \xrightarrow{\substack{-\frac{1}{3}r_2 \rightarrow r_2 \\ -\frac{1}{6}r_2 \rightarrow r_3}} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & \frac{5}{3} \\ 0 & 1 & \frac{5}{3} \end{bmatrix} \xrightarrow{-r_2+r_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & \frac{5}{3} \\ 0 & 0 & 0 \end{bmatrix}$$

2 leading 1's $\rightarrow \text{rank}(A) = 2$

1 free variable $\rightarrow \text{nullity}(A) = 1$

or $\text{nullity}(A) = \# \text{ of columns} - \text{rank}(A)$
 $= 3 - 2 = 1.$

(b) Find c if the dimension of the row space of C is 2.

$\rightarrow C$ is 3×3 , $\text{rank}(C) = 2$
 $\Rightarrow \det(C) = 0.$

$$\det(C) = \det \begin{bmatrix} 1 & 0 & 2 \\ (c^2-2) & 0 & 4 \\ 2 & 1 & 3 \end{bmatrix} = -1(1 \cdot 4 - (c^2-2)(2)) = 0$$

$$4 - 2c^2 + 4 = 8 - 2c^2 = 0$$

$$2(4 - c^2) = 0$$

$$C = \pm 2$$

Why does rank not drop to 1?
 $c = \pm 2$

$$\hookrightarrow \begin{bmatrix} 1 & 0 & 2 \\ 2 & 0 & 4 \\ 2 & 1 & 3 \end{bmatrix}$$

not scalar mult. of each other,
 so they are linearly indep.

4. (5+5 points)

5 rows, 4 columns

(a) Let D be a 5×4 matrix whose rank is 3. Is it possible that

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 0 \end{bmatrix} \text{ and } \mathbf{x}_2 = \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

are both solutions of $D\mathbf{x} = \mathbf{0}$? State yes or no, and justify your answer.

$\vec{x}_1$ and $\vec{x}_2$ both live in the nullspace of D , and they are linearly independent $\Rightarrow \underbrace{\dim \text{nullspace}(D)}_{\text{nullity}(D)} \geq 2$.

$\text{rank}(D) + \text{nullity}(D) = 4 = \# \text{ of columns}$
if $\text{rank}(D) = 3$,

$$3 + \text{nullity}(D) = 4 \Rightarrow \text{nullity}(D) = 1. \quad \text{contradiction.}$$

No, this is not possible.

(b) Let W be a subspace of P_3 , where

$$W = \{at^3 + bt^2 - at + b \mid a, b \in \mathbb{R}\}.$$

i. Find a basis for W .

ii. What is $\dim W$?

$\textcircled{i} \quad a(t^3 - t) + b(t^2 + 1)$
basis for W : $\{t^3 - t, t^2 + 1\}$.
span W , and we lin. indep. b/c they're not scalar mult. of each other.

$\textcircled{ii}$ basis has 2 elements, so

$$\dim W = 2.$$

5. (4+4 points) Make sure to justify your answer to the following questions.

(a) Let $L: P_2 \rightarrow P_3$ be a function defined by

$$L(at^2 + bt + c) = (a+c)t^3 - b + 1.$$

Is L a linear transformation?

No. Consider $t^2 + t + 1$, $r=4$.

$$\begin{aligned} L(4(t^2 + t + 1)) &= L(4t^2 + 4t + 4) \\ &= (4+4)t^3 - 4 + 1 = 8t^3 - 3 \end{aligned}$$

vs.

$$\begin{aligned} 4L(t^2 + t + 1) &= 4((1+1)t^3 - 1 + 1) = 4(2t^3) \\ &= 8t^3 \end{aligned}$$

not the same.

(b) Let $K: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a function defined by

$$K\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} 2u_2 - u_3 \\ u_3 + 3u_1 \end{bmatrix}$$

Is K a linear transformation?

Yes!

Option A: $\vec{u}, \vec{v} \in \mathbb{R}^3$
 $c \in \mathbb{R}$.

$$\begin{aligned} K(\vec{u} + \vec{v}) &= K\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = K\left(\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}\right) \\ &= \begin{bmatrix} 2(u_2 + v_2) - (u_3 + v_3) \\ u_3 + v_3 + 3(u_1 + v_1) \end{bmatrix} = \begin{bmatrix} 2u_2 + 2v_2 - u_3 - v_3 \\ u_3 + 3u_1 + v_3 + 3v_1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 2u_2 - u_3 \\ u_3 + 3u_1 \end{bmatrix} + \begin{bmatrix} 2v_2 - v_3 \\ v_3 + 3v_1 \end{bmatrix} = K\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) + K\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = K(\vec{u}) + K(\vec{v})$$

$$K(c\vec{u}) = K\left(c\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = K\left(\begin{bmatrix} cu_1 \\ cu_2 \\ cu_3 \end{bmatrix}\right) = \begin{bmatrix} 2cu_2 - cu_3 \\ cu_3 + 3cu_1 \end{bmatrix} = c \begin{bmatrix} 2u_2 - u_3 \\ u_3 + 3u_1 \end{bmatrix}$$

Option B:

$$\begin{bmatrix} 0 & 2 & -1 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 2u_2 - u_3 \\ u_3 + 3u_1 \end{bmatrix} = c \begin{bmatrix} 2u_2 - u_3 \\ u_3 + 3u_1 \end{bmatrix} = c K\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = c K(\vec{u})$$

Yes, b/c K is a matrix transformation.

6. (5+5 points)

(a) Let W be the following subset of $\mathbb{R}^3$:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid xz = 1 \right\}.$$

Is W a subspace of $\mathbb{R}^3$ with the usual vector addition and scalar multiplication?

No. Consider $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \in W$.

$4 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix}$ is not in W , since $4 \cdot 4 = 16 \neq 1$. Not closed under scalar mult.

Could also do: $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ in W , but

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \leftarrow \text{not in } W, \text{ since } 2 \cdot 2 = 4 \neq 1.$$

(b) Let P_* be a subset of P defined as follows:

$$P_* = \{ a_2 t^2 + a_1 t + a_0 \mid a_0, a_1, a_2 \in \mathbb{R} \text{ and } a_0 + a_2 = 0 \}.$$

Is P_* a vector space with the usual polynomial addition and scalar multiplication?

Nonempty? $0t^2 + 0t + 0 \in P_*$ $\Rightarrow 0+0=0 \checkmark$.

Closed under +: let $p = a_2 t^2 + a_1 t + a_0 \in P_*$, $q = b_2 t^2 + b_1 t + b_0 \in P_*$, $a_0 + a_2 = 0$, $b_0 + b_2 = 0$

$$\begin{aligned} p+q &= (a_2 t^2 + a_1 t + a_0) + (b_2 t^2 + b_1 t + b_0) \\ &= (a_2 + b_2) t^2 + (a_1 + b_1) t + (a_0 + b_0) \end{aligned}$$

$\leadsto$ check: $(a_0 + b_0) + (a_2 + b_2) = (a_0 + a_2) + (b_0 + b_2) = 0 + 0 = 0$.

Closed under scalar mult: let $p \in P_*$, $c \in \mathbb{R}$

$$cp = c(a_2 t^2 + a_1 t + a_0) = ca_2 t^2 + ca_1 t + ca_0 \leadsto ca_2 + ca_0 = c(a_2 + a_0) = c \cdot 0 = 0.$$

so $cp \in P_*$.

Yes! This is a subspace⁸ of P_* , so it's a vector space, as well!